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SUB WORKING GROUP 3

Use of low-cost devices for measuring acceleration onboard trains

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Abstract

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DISCLAIMER

The Harmotrack project is a UIC project managed by SNCF Réseau, with many other members, railway companies, infrastructure managers and academic institutions contributing to the project. More than 60 companies took part, using their own funds, and based on their own technical issues.

Whilst efforts have been made to ensure the accuracy and completeness of this document, the Harmotrack project leaders shall not be liable for any errors or omissions, however caused.

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ABSTRACT

The purpose of this technical report is to collect studies, developments, and general information on the use of low-cost devices for measuring acceleration onboard trains.

The report initially presents the main systems treated, then data collection and processing are defined alongside useful information for track monitoring. Different techniques to improve track localisation are explained before a validation process for a specific measurement system is described. Finally, the impact that an accelerometer's position has is evaluated.

TERMS, ACRONYMS AND ABBREVIATIONS

ACRONYM	DESCRIPTION
ABA	Axle box acceleration
C95	95th percentile
GNSS	Global navigation satellite systems
GPS	Global positioning system
IMU	Inertial measurement unit
PK	Kilometric point
PSD	Power spectrum density
RMS	Root mean square
SNCF Réseau	French infrastructure manager
SWG	Sub-working group
UIC	International Union of Railways

INTRODUCTION

The UIC Harmotrack project aims to explore all of the possible uses of dynamic measurements (either acceleration or force) for railway track monitoring. Started in 2019, the project was divided into three sub working groups (SWG) in 2021 once the technical goals had been established.

Sub-Working Group 3 of the Harmotrack project was dedicated to identifying measurement devices that have a low investment cost but still offer sufficient precision for track maintenance needs. With electronic components becoming smaller and nevertheless achieving higher accuracies, it is now possible to have accurate acceleration sensors for onboard train measurements through smartphones and cheap setups. The group focused on different aspects of how to implement these solutions. From smartphone setup and requirements to data validation and processing, this report aims to guide users in implementing these systems on their network.

This SWG tackles two of the Harmotrack project's main objectives, which were to set a benchmark on measurement techniques for dynamic responses and to define the scope of application for the acceleration measurements as well as best device specifications. This report responds to these two objectives through a spectrum of reduced costs and easily accessible equipment.

The report is divided into six chapters:

1) Definition of the measurement systems
2) Data collection
3) Data processing
4) On-track localisation
5) Validation of the low-cost devices
6) Impact of the accelerometer position

1. DEFINITION OF THE MEASUREMENT SYSTEMS

Accelerometers are cheap pieces of equipment compared to geometry measurement systems which require sophisticated lasers to measure millimetric rail deviations. They deliver useful information on the quality of the track geometry, despite not providing an exact evaluation of the geometry defects when measured inside of the carbody. Therefore, it is useful for infrastructure managers to have the acceleration measurements from their tracks in order to have comprehensive knowledge of the tracks' quality, as well as an affordable means of using a railway vehicle to measure the impact of a defect.

Chapter 1 describes the different types of low-cost devices that have been studied and presented by infrastructure managers in the SWG3. From smartphones to custom-made measurement systems, it shows different approaches when it comes to measuring acceleration aboard a train carbody.

1.1. SMARTPHONES

Embedded sensor technology in smartphones is constantly improving and its performance to size ratio is impressive. Using smartphones is practical, as they have already been deployed and used amongst the maintenance teams, and make acceleration measurements more flexible and less expensive.

The embedded sensors usually included in a smartphone are listed below:

- **Accelerometer** - acceleration in m/s2 in the X, Y, and Z directions, from an accelerometer on an Android™ device.
- **Gyroscope** - the rate of rotation in **rad/s** around the X, Y, and Z axes, from the gyroscope on an Android™ device.
- **Light Sensor-** the ambient light in **lux (lx)**, from the light sensor on an Android™ device.
- **Temperature Sensor**- the temperature in °**C**, from the temperature sensor on an Android™ device.
- **Pressure Sensor -** the atmospheric pressure in **hectopascals (hPa)** , from a pressure sensor on an Android™ device.
- **Humidity sensor -** the relative ambient air humidity as a **percentage (%)**, from a humidity sensor on an Android™ device.
- **Location Sensor -** the latitude, longitude, and altitude, from a GPS receiver on an Android™ device.

- **Orientation -** the rotation of the device, using three angular quantity vectors – pitch (around the X axis), roll (around the Y axis), and azimuth (around the Z axis). The outputs are the angles as a 1-by-3 vector **in degrees**.
- **Magnetometer -** the strength of the magnetic field around an Android™ device. The built-in magnetometer sensor measures the magnetic field along the X, Y, and Z axes. The block outputs the magnetic field as a 1-by-3 vector in **microtesla (µT)**.

Specification datasheets from the internal sensors of recent smartphones often display the following behaviour, specific to acceleration measurements:

- Acceleration ranges from ±2g up to ±16g
- Bandwidth ranges from 100 to 500Hz
- Output rate ranges from 20 to 2000Hz

These specifications show the capability of smartphone sensors to accurately measure accelerations within a train carbody as the desired frequencies are usually within the 0-20Hz range. However, using smartphones would not be possible on axle boxes and bogies due to the lack of high frequency measurements. In order to use smartphones to replace conventional measurement systems inside train carbodies, a validation process is presented in Chapter 5 “Validation of low-cost devices”.

Some applications already exist on smartphone app stores for recording data coming from the phone’s internal sensors. Additionally, specific train track interaction measurements were developed within the study group, which are presented in the Chapter 2 “Data collection”.

1.2. LOW-COST INSTRUMENTS

1.2.1. INFRAESTRUTURAS DE PORTUGAL

Other acceleration measurement devices can be used at a very low cost such as instruments combining a microchip and an accelerometer. A setup used by Infraestruturas de Portugal is presented in Figure 1.

An ESP32 development board was combined with a MPU6050. The MPU6050 has 6 degrees of freedom and can measure with a maximum range of +/-1g. This setup is then connected to a computer for monitoring the measured data.

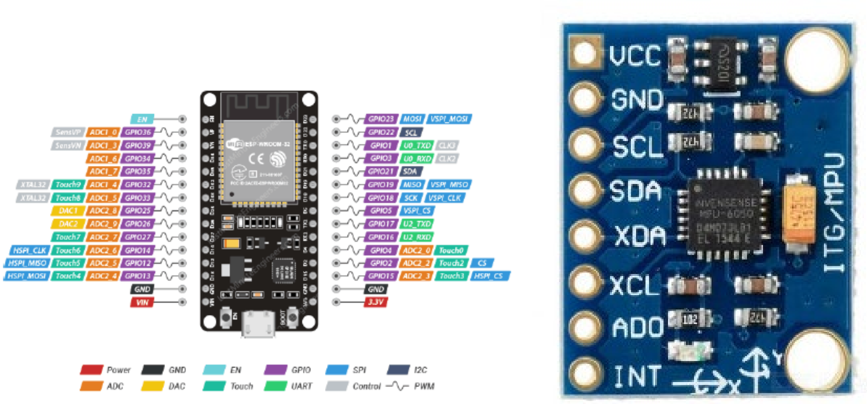


Figure 1: Infraestruturas de Portugal's measurement system

These devices can be used for measuring acceleration and rotation for non-sensitive applications. However, this type of device cannot be used for axle box measurements as the maximum acceleration range is too short.

The advantage of this setup is that the cost is very low, and the sampling rate can be adjusted to relatively high values (up to 1kHz).

1.2.2. RUMO LOGISTICA

The Brazilian railways do not yet have standards regarding the measurement of acceleration for track maintenance, the standards are focused on measuring geometry through inspection with measurement trains.

Nevertheless, the use of acceleration measurements is still being explored by some companies and university institutions. The study for this project was conducted at Rumo Logistica, a company located in the south of Brazil. Freight and passenger trains operate on the tracks used for this study.

The assembled sensor is called VibMapper and has the following features:

Technical Data:

- Accelerometer: Mpu6050
- Micro controller: Esp8266
- GPS: WJ Neo-6m V2
- Storage: Micro SD
- Sensor protection case material: ABS thermoplastic polymer
- The following accelerations are measured at a sampling frequency of 200Hz:
- Vertical axle box acceleration

- Vertical bogie acceleration
- Lateral bogie acceleration
- Vertical carbody acceleration
- Lateral carbody acceleration

The location of the first measurements over 120 kilometres were very precise. For this measurement, the sensor was placed on the floor of a locomotive. The aim of the project was to validate the data using a geometry measurement car and then monitor the infrastructure with commercial trains equipped with accelerometers (see Appendix 1: Rumo Logistica's device, for more details).

1.3. REFERENCE SYSTEMS

The reference systems were used as baseline measurements for the different low-cost devices. Regardless if these were dedicated measurement trains equipped with accelerometers such as IRIS 320 (SNCF Réseau) or PIRIS Reis (TCDD), or even specific portable measurement devices such as the accelerometric suitcase (see *Appendix 2*: Data References*,* for a more extensive overview of the reference systems), these measurement systems can be used to validate the low-cost devices.

When a low-cost device is initially introduced into a network, a comparison with current measurement systems should be carried out in order to ensure that the low-cost devices are sufficiently reliable.

1.4. SUMMARY

In summary, this chapter explored the type of measurement systems which may be used by different infrastructure managers, and gave an overview of the future possibilities for low-cost devices for measuring accelerations.

2. DATA COLLECTION

This chapter focuses on the implementation of low-cost devices onboard trains. It gathers information from different tests to describe recommendations for the use of smartphones for track quality measurement.

2.1. GENERAL CONSIDERATIONS

In order to detect track defects, data is collected by installing the low-cost device on the floor of the train. When assessing passenger comfort, the sensors could also be placed either on the floor or on a seat/table to assess the acceleration felt by a customer. However, this use-case is not the focus of the UIC Harmotrack project.

Where possible, the sensors should be attached to the floor to limit them sliding in curves and during brake applications. This also allows the defects to be better detected and reduces the impact of the support structure (seat, table, other) in detecting peaks.

The measurement of track defects with low-cost devices requires at least three components:

- an acceleration sensor
- a GPS or speed sensor
- a chip or computer to save the data

In order to improve location tracking, the GPS data sampling rate should be at least 1Hz. For an improved and more practical use of the measured data, a GPS sampling rate of 10Hz is required, which means one point every 8 meters with a running speed of 300km/h.

An issue with collecting GPS data is that it can lose signal due to speeds, tunnels, high density urban areas, or even because of how the glass of the train windows has been treated.

There are different solutions to this:

- Using a more accurate and developed GPS sensor (some Bluetooth sensors can be coupled with smartphones to improve their location services, the KELI app developed by SNCF Réseau is compatible with external sensors)
- Using a speed estimation method based on accelerometric data
- Using speed estimation via two-point measurements

The latter two methods are described in more detail later on in this document.

Using an external GPS sensor can improve both the sampling rate and the signal reception, thereby improving the peaks' location data for the maintenance team. A wireless GPS sensor can be put on the side window of a train to acquire better signal.

The acceleration sensor should, at least, measure the vertical and lateral components of the acceleration (requiring a 2-axis sensor). However, for improved data use and for different post-processing calculations, it is recommended to have a 3-axis sensor with a gyroscope measuring angular speed (3-axis) and position. These measurements can then be used to calibrate the sensor to eliminate gravity and the effect of curves and ramps through sensor fusion.

Certain sensors integrated in smartphones or used as part of an industrial setup have extensive information such as linear acceleration or gravity values under the 3 axis of the sensor, which are pre-programmed outputs. However, it is important to check that these outputs contain the frequency needed for analysis. For example, certain methods simply apply a high pass filter to eliminate gravity, but this can also eliminate accelerations due to ramps or curves in the signal. Depending on how the accelerometers are used, this should not be neglected when dealing with pre-programmed outputs from Android, iOS, or others.

2.2. KELI – SNCF RÉSEAU'S APPLICATION FOR TRACK MONITORING

SNCF Réseau has developed a smartphone app to respond to the need for a measurement application dedicated to track monitoring (Figure 2). This app is called KELI and uses internal accelerometers to measure a train's dynamic behaviour and a GPS sensor to deduce the train's speed and location.

The start screen allows the user to enter general information about the journey, including the line, track, the initial kilometric point, and train type. A measurement screen then displays three graphs for vertical (AVC), lateral (ATC) and longitudinal (ALC) accelerations, as well as the current location and speed of the train. The angular positions of the smartphone are also stored during the run.

Only raw data is stored during the recording, as it was observed that filtering and post-processing the data directly during the measurement potentially disturbed the smartphone and caused lagging, data loss or even closed the app. The filtering process and any thresholds exceeded are therefore calculated in a post-processing tool.

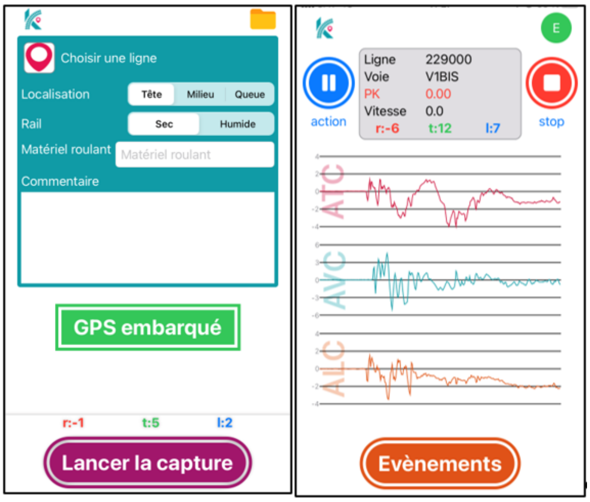


Figure 2: The KELI app

2.3. OTHER COMMERCIAL APPLICATIONS

Other apps can be separated into two groups according to the way they work:

1. Collect data, apply filters (depending on the application) with storage in its own memory, and visualise it in the graph. In this group, the smartphone works as both a sensor and acquisition unit. This group has some limitations:
   - The sampling rate can be reduced because of high ram usage
   - Complex visualisation increases ram usage
   - Real time monitoring is not possible
2. The phone memory is used to record the data Collect a data stream from another device via a local ethernet, bluetooth or internet connection. In this group, the smartphone acts as a data acquisition system, with the sensor sending raw data to the smartphone. The disadvantage of this is needing another device for collecting and visualising the data, as well as the risk of losing the connection between the sensor and the smartphone.

2.4. SUMMARY

The minimum requirements for setting up low-cost devices to monitor track quality have been established in this chapter, with different issues and focal points being raised and solutions being discussed. KELI, the tool developed by SNCF Réseau is an example of a dedicated app for track monitoring with a solution implemented for measuring accelerations and determining locations.

3. DATA PROCESSING

The data collected by smartphones through the KELI app is raw acceleration data that needs to be post-processed to make it usable.

SNCF Réseau has developed a Python algorithm to carry out different steps to process the data.

3.1. DATA CORRECTION

The data correction process is described in the Figure 3.

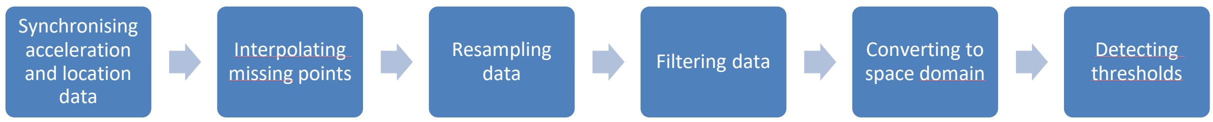


Figure 3: Data Processing

The first step consists of synchronising the acceleration and location data as they come from different sources inside the smartphone. The data collected generated two different datasheets, one for the accelerometer and one for the GPS sensor. They have different sampling rates, and are synchronised by their timestamps.

As acceleration data is collected more frequently than GPS data, there are missing points for location data, which are complemented by simply interpolating the GPS coordinates and the kilometric points with regards to the timestamp.

It has often been observed that smartphone sensors do not produce data at an exact frequency, and that the timestamp can deviate after longer measuring times. In order to solve this issue and to be able to apply a filter to the dataset, this step consists of resampling the data to an exact chosen frequency to avoid issues when using the data (especially when applying a filter).

The filtering process is described in the next section of this document.

After the data has been filtered, the algorithm converts time domain data into space domain data by resampling the data to a fixed spatial sampling rate of 0.25m, for this to match the data coming from track measurement trains and devices, and to simplify the use of the measurements for the maintenance team.

The final step in the process is to detect any exceeded thresholds with a specific algorithm using SNCF Réseau's internal standards. This allows the maintenance team to identify the points where monitoring or repair is needed and provides information about the type of defect to search for.

3.2. FILTERING

The filters applied to the measured data are currently taken from the European standard *EN 14363*. The purpose of measuring carbody accelerations is to assess the track quality and evolution through regular measurement campaigns.

3.3. SUMMARY

Similar processing to that described above can be used to deal with measurement data from low-cost devices. The data correction process must be thorough before the signal is filtered and the bandwidths of interest are chosen. The common applications for this data are to determine ride comfort or analyse track defects. For the latter, cut-off frequencies and thresholds can be taken from *EN 14363*.

4. ON-TRACK LOCALISATION

The purpose of measuring acceleration onboard commercial trains is, as defined by this sub working group, mainly to evaluate track quality and detect track geometry defects. Therefore, the peak values which exceed certain thresholds or evolve rapidly should have a precise location for maintenance teams to identify and correct the defects. However, due to the train speed, the lack of data on the train's actual speed and position, and difficulties with having a GPS signal inside the train carbody, there can be discrepancies in the position. This is especially the case in tunnels or dense areas, and therefore complementary methods must be implemented.

Different methods have been developed by the members of the SWG3 to more precisely locate the measurement points. These methods are used first to assess the speed of the vehicle, and then deduce the current position on a specific line and track.

4.1. EXTERNAL GNSS RECEIVERS AND MEASUREMENT LOCATION

To increase the GPS accuracy and to reduce losing signal, the first solution is to connect an external GNSS receiver. Many external receivers exist and can be connected via Bluetooth or cable to low-cost devices. External GNSS sensors also have a higher output frequency than smartphones (10Hz vs 1Hz), as well as better signal reception and can be put in a separate location to the smartphone.

Moreover, due to the train carbody and the treatment of the windows, Bluetooth, WIFI and GNSS signals are often filtered, resulting in having low reception inside the carbody. Attaching a GNSS receiver either onto the window (inside the train) or if possible, outside the carbody on top of the train greatly improves the signal and therefore the location of the measurement points.

Additionally, it can be useful for the maintenance team to register certain occurrences during travel such as passing over a switch, a bridge, or through a tunnel, as these can also help locate a defect with precision.

4.2. ESTIMATING SPEED USING FREQUENCIES IN THE ACCELEROMETRIC DATA

This section describes the use of resonance frequencies, or harmonic content, on a passenger train to estimate vehicle speed. The method detailed here is based on previous research already published in Vehicle System Dynamics [1].

The tests were carried out on a high-speed line. Data for the speed estimation process were acquired from an in-service passenger train, equipped with sensors on the axle box (A), bogie (B) and in the cab (C). The test device used was based on a previous development by BCRRE [2, 3]. The leading axle boxes were equipped with vertically sensing accelerometers, acquiring data at a sample rate of 2500 samples per second. The bogie and the cab were each equipped with a GNSS receiver measuring the latitude and longitude, speed, and heading at 4Hz, and an IMU with accelerometers and gyroscopes acquiring data on each direction at 250Hz. The IMU reference system with respect to the vehicle is presented in Figure 4.

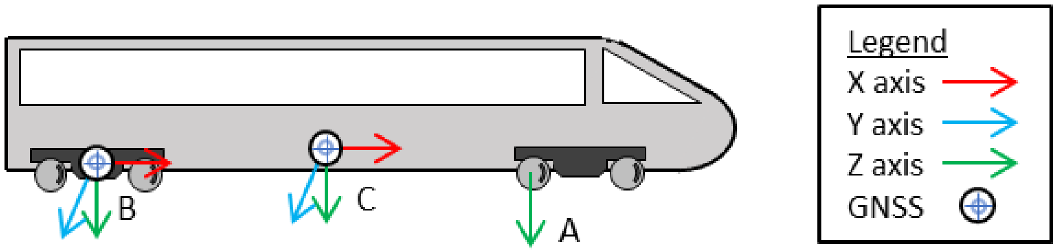


Figure 4: IMU axis system directions

No other sources of information from the train were provided for the measurements, in particular the speed data from the train tachometer was not available.

Using sensors onboard passenger trains, a relationship between the vehicle speed and the frequency content of the vehicle was detected, as observed in Figure 5. The GNSS estimated speed and the spectrograms of the axle box sensors can be compared, with there being several harmonics showing a correlation to speed, and slight differences in the frequency spectrum in tunnels (area shaded in grey).

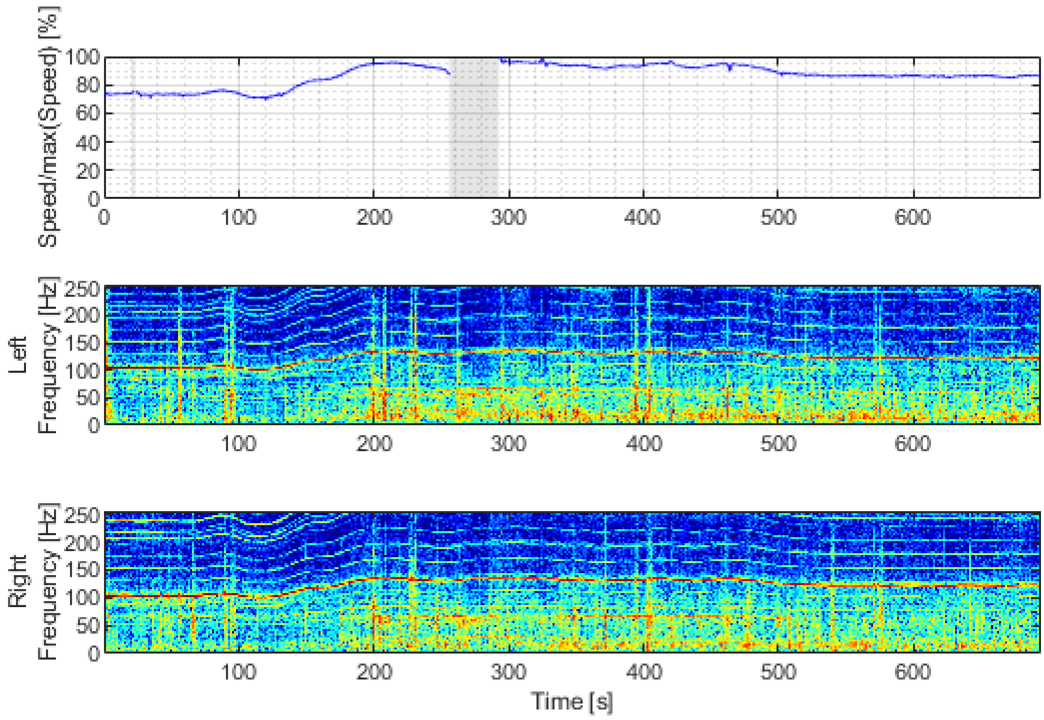


Figure 5: Comparison between GNSS speed estimation and axle box spectrograms

The speed estimates are extracted by obtaining the highest frequency with a peak in the magnitude of the spectrogram and applying a low-pass filter to the signal to smooth the result. Figure 6 shows the speed estimates for each of the vertical accelerometers. It was observed that the axle box and the bogie estimates were more reliable than the GNSS speed estimates, which is most likely due to the damping effect of the vehicle's secondary suspension, as it filters out most of the frequency content of the vibrations above a certain frequency.

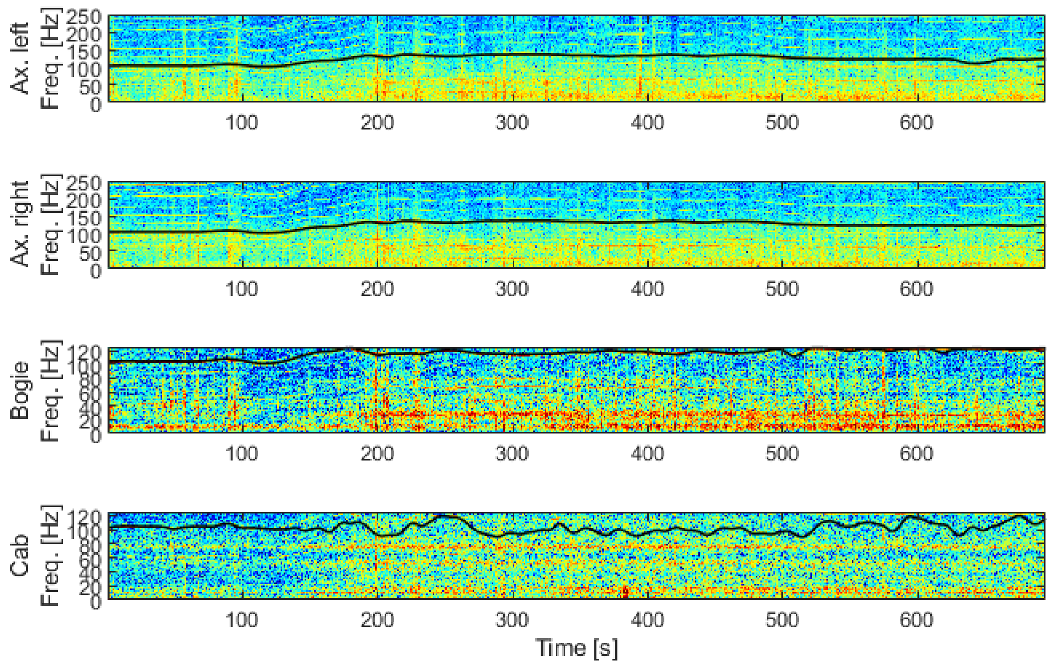


Figure 6: Curve fitting for selected signals

Due to its low signal-to-noise ratio, the cab data was not used in the speed estimates. Once the estimates for each signal were obtained, the ratio between the harmonics and speed was calculated dividing the values of the harmonics by the speed curve at each sampling point, and then averaging the values.

The estimates were then combined using a Kalman filter [4], using the bogie GNSS speed, cab GNSS speed, and estimates from the harmonics as measurements. The longitudinal acceleration measured at the bogie and in the cab were considered as correction factors using the acceleration to obtain the change of speed over time. Figure 7 shows the speed estimate from each source, and Figure 8 shows the error in respect to the bogie GNSS. As the segments without GNSS signal did not have a speed estimate, no error was calculated for these.

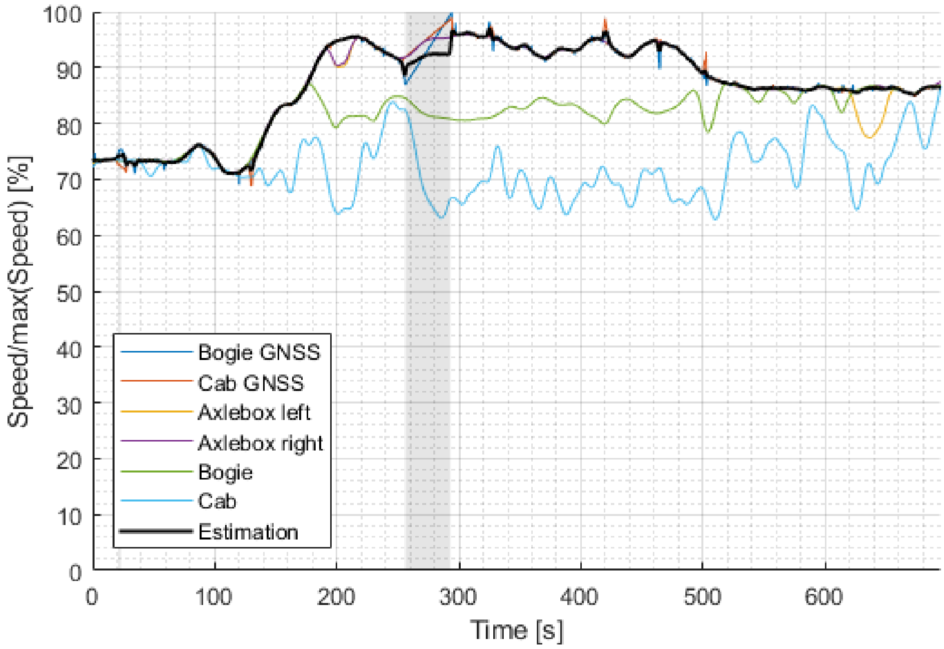


Figure 7: Speed estimation comparison

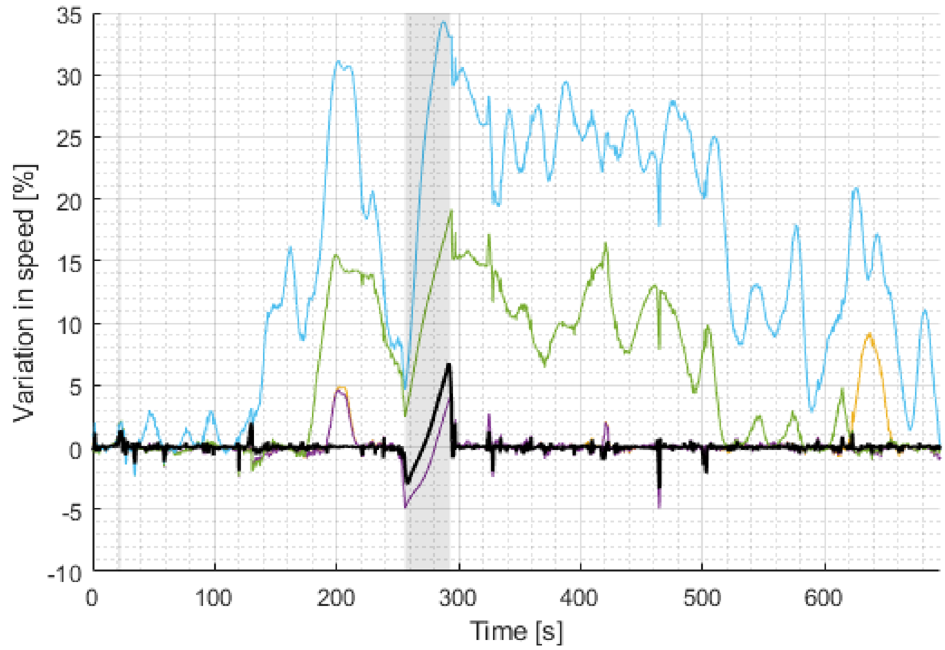


Figure 8: Comparison of speed estimate errors

The errors in the estimates was generally observed to be below 2%, and even lower with the Kalman filter, using the bogie GNSS estimates for reference. Both of the axle box sensors were the most accurate, whilst the bogie estimate had errors of up to 15%, and up to 35% for the cab. This is most likely due to the damping effect of both the primary and secondary suspension. In a tunnel, the error cannot be determined without GNSS, however the speed estimates can be used, as the magnitude of speed is similar both inside and outside of a tunnel.

4.3. ESTIMATING SPEED USING ACCELEROMETRIC DATA AT TWO POINTS

This section describes the use of two accelerometers mounted on different axle boxes to calculate vehicle speed. With this method, the speed is calculated by observing the time difference between two acceleration sensors placed at different points on the vehicle. The method has been detailed and tested with the dataset shown below in Figure 9.

The first accelerometer is placed on the first axle box (point A) and the other is placed on the last axle box (point B) of the train. The train passes on a test track with a longitudinal defect, the accelerometers collect data while the train is moving.

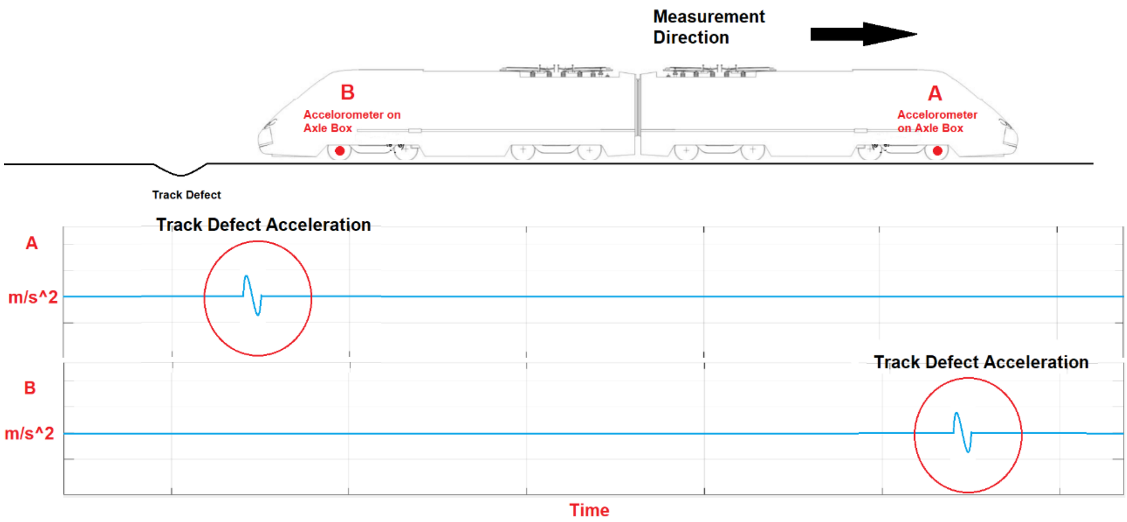


Figure 9: Illustration of the speed estimation method

As the train passes over the track defect, high acceleration is seen in the sensor data at point A in the direction of movement, and after a while, this high acceleration is also seen at point B. The time elapsed between these two high accelerations is the time taken to travel the distance between point A and B. The distance between point A and point B is constant and thereby when the distance is divided by the time difference, the speed is obtained. The principle is shown in Figure 10.

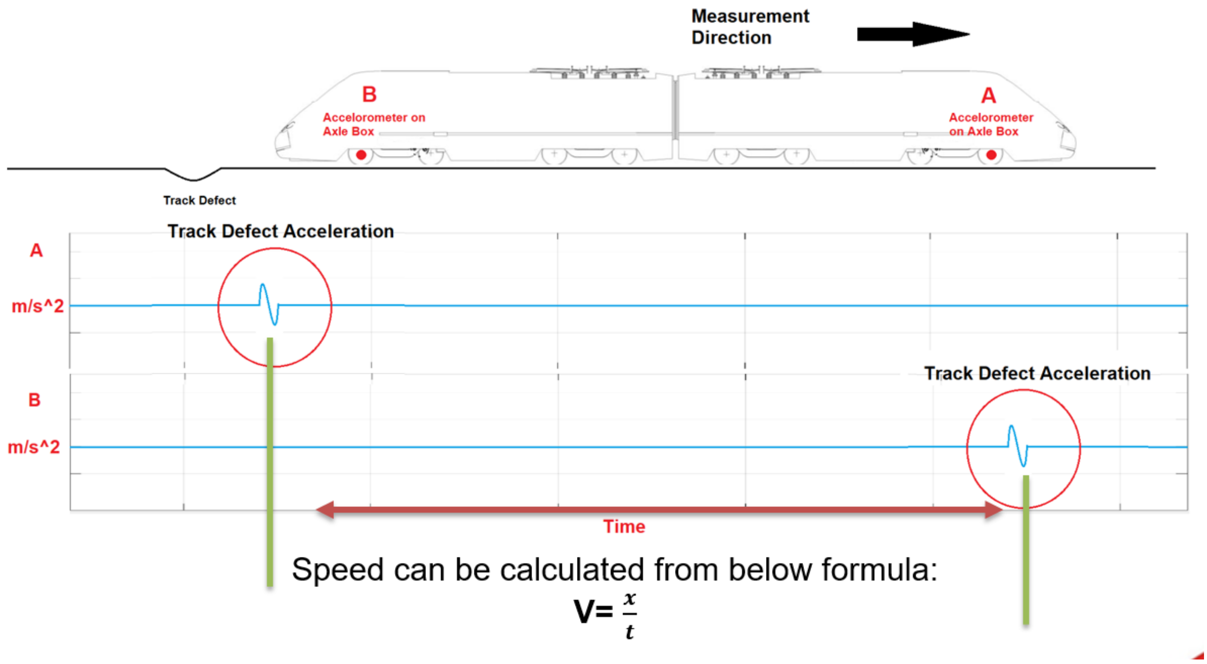


Figure 10: Speed calculation from the time difference

This method was applied to the TCDD Piri Reis Measurement Train's dynamic measurement data to determine the accuracy of the method. Two accelerometers were selected with a distance between them of 19 meters.

The data measured from these two sensors is shown in Figure 11, where there is a visible similarity between the two signals, but with a time difference due to the different sensor positions. The method can be applied to detect vehicle speeds at any occasion.

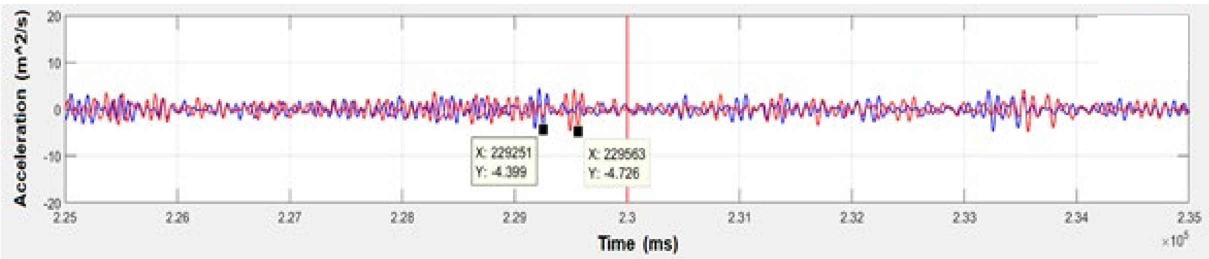


Figure 11: Accelerometric data

Figure 12 shows the comparison between the speed measured by the train tachometer and the estimated speed from the described method.

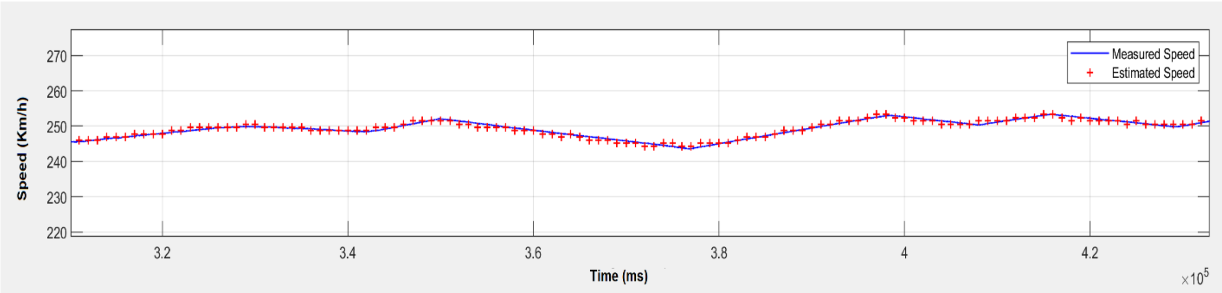


Figure 12: Measured speed and calculated speed

The calculated speed is well in line with the measured speed. The accuracy of the calculated speed depends on the vehicle speed and the distance between two acceleration sensors. When the distance increases, the calculated speed accuracy increases but the accuracy in the speed changes decreases.

As a result, with the method described above, acceleration sensors can be used to calculate the speed with high precision without installing extra equipment on the vehicle.

4.4. SUMMARY

Where a measurement was taken is critical for on-track maintenance work. The three approaches described in this chapter provide specific techniques to locate the measurement's location after having been collected. In general, a combination of different methods and data sources is useful for ensuring the reliability of the position.

5. VALIDATION OF THE LOW-COST DEVICES

The validation process used in this chapter is based both on European standards (EN 13848) for track geometry measurements and from SNCF Réseau's internal standards (PSIGT.EV.315). The European standards only provide specifications for validating geometry measurements, but the general principles can be applied to acceleration. The SNCF Réseau standards have been developed using the *EN 13848* standards, internal knowledge and experience on accelerometers.

The validation process defined at SNCF Réseau for the high-speed measurement train IRIS 320 requires specific runs on an identified track. It involves three types of validation:

- Repeatability: same conditions (speed, orientation, direction)
- Reproducibility: different orientation
- Comparability: different sensors

5.1. STATISTICAL ANALYSIS

Statistical analysis requires both signals to be perfectly being synchronised before being compared. The signals should be in the space domain; however, an equivalent comparison can be conducted with time domain signals.

The first step is to calculate the point-to-point difference between the synchronised signals. The distribution of these differences can be evaluated to determine the system's precision.

An often-used metric is C95 (95th percentile). A C95 of 0.2m/s² can be defined as 95% of the absolute differences observed being lower than 0.2m/s².

5.2. FREQUENTIAL ANALYSIS

Frequential analysis consists of analysing the PSD (power spectrum density) of two signals. This analysis can be conducted with previously observed signals to ensure that the gain and filtering frequency are still valid.

For comparability, reproducibility or repeatability, the ratio of the PSD of signal 1 divided by the PSD of signal 2 should be within a certain range, around 1 for the spectrum of interest (in the space domain or time domain).

For carbody accelerations, the wavelength of interest for track maintenance that is considered at SNCF Réseau is between 10 and 150 meters.

5.3. VALIDATING THE IPHONE 8

The validation of acceleration measurements using an iPhone 8 used windows of 1 min (distance of 5km at 300km/h) to evaluate the standard deviation of both signal, with the portable measurement system being the reference system used. The data recorded by the iPhone 8s were acquired through the KELI app (see 2.2) and the comparison was carried out on filtered signals.

Statistical and frequential analysis (see 5.1 and 5.2) were conducted on both vertical and lateral accelerations. The results were compared with the tolerances defined in SNCF Réseau standards.

During the test, all the measurement systems were placed inside a coach aboard IRIS 320. For the comparison, the line’s maximum speed of 300km/h was used.

Figure 13 and Figure 14 show a comparison of the vertical and lateral acceleration of the accelerometric suitcase and the smartphone signals in the time domain.

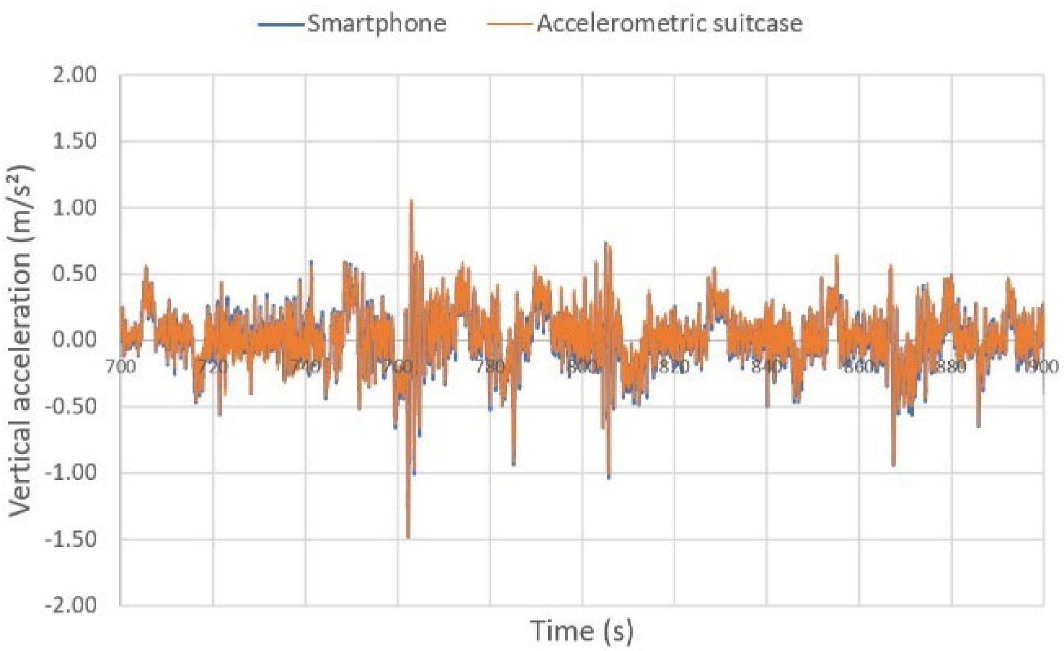


Figure 13: Vertical acceleration – superimposed signal

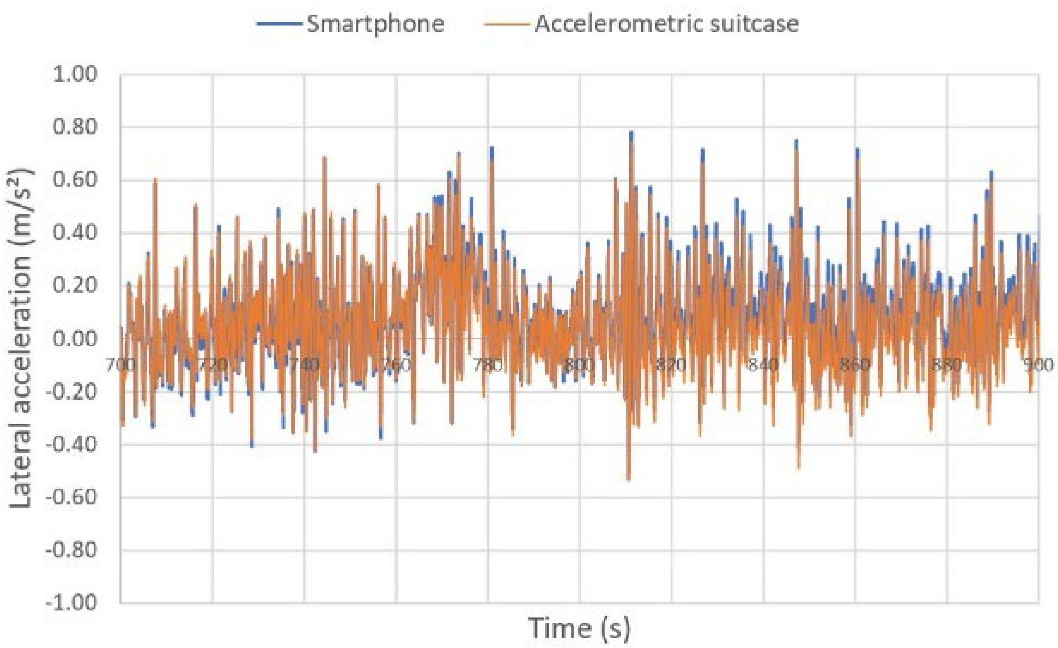


Figure 14: Lateral acceleration – superimposed signal

The acceleration observed in both signals are very close. The differences are evaluated as well as their distribution and presented in Table 1.

Table 1: Differences between the reference system and the iPhone 8

Acceleration	Difference in acceleration values (m/s²)		Difference in standard deviation (m/s²)		Cumulative measured distance (km)
	Max	C95	Max	C95	
Vertical	0.34	0.12	0.017	0.012	325
Lateral	0.55	0.13	0.017	0.014	

The results are compliant with the SNCF Réseau standards. They show that the embedded acceleration sensor of the iPhone 8 works wells and is reliable.

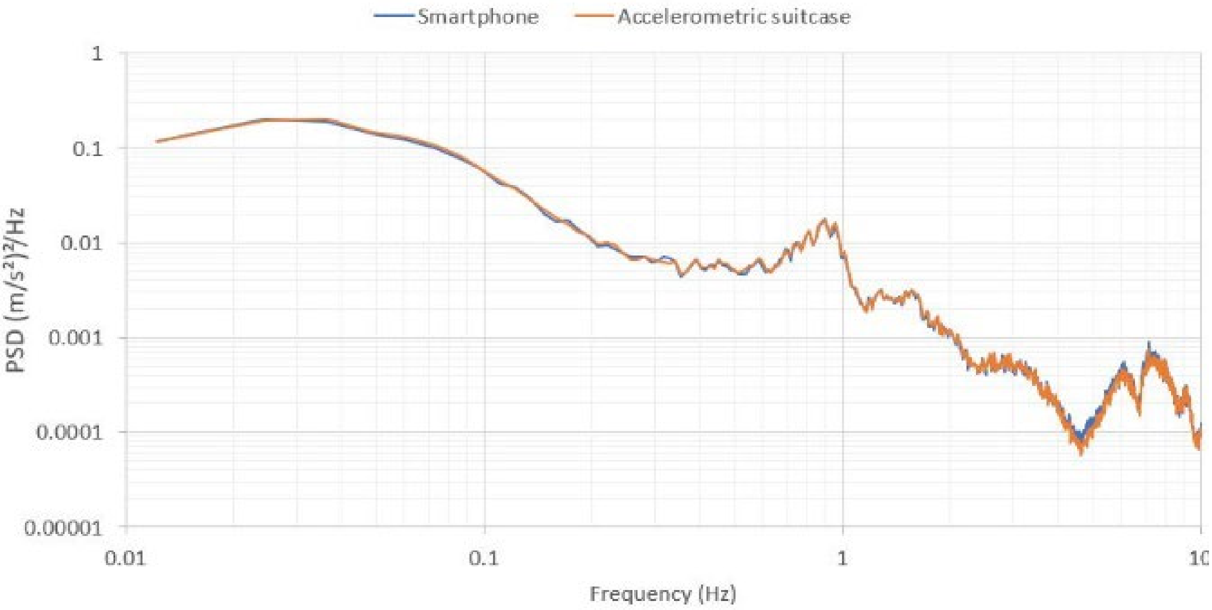


Figure 15: Vertical acceleration – superimposed PSD

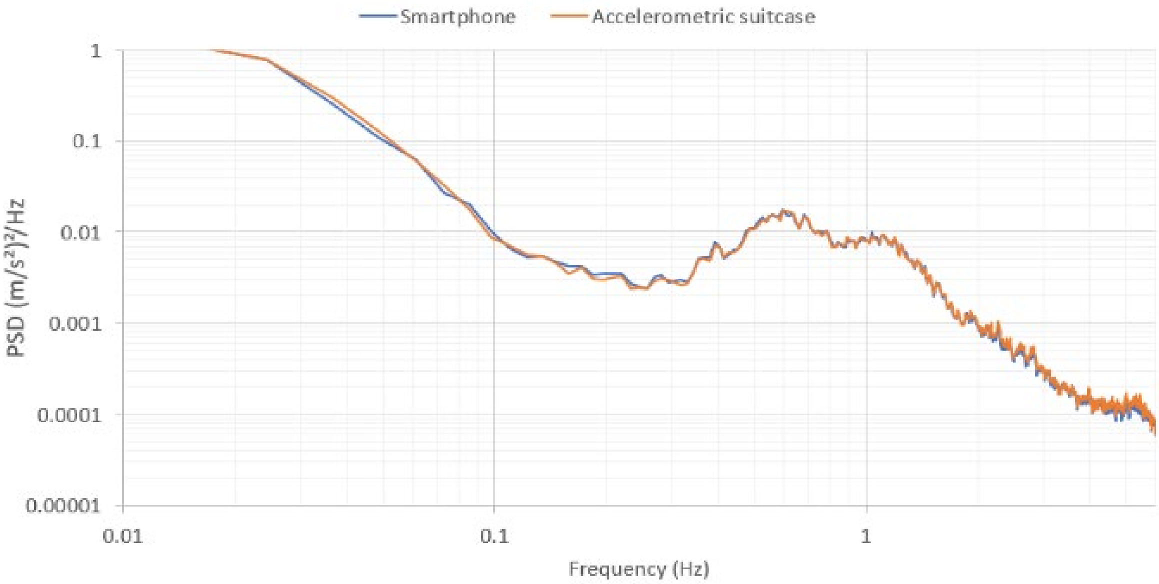


Figure 16 : Lateral acceleration – superimposed PSD

The PSD of both signals are shown in Figure 15 and Figure 16. Both graphs are superimposed, which shows similar energy levels in the spectrum of interest (here in the time domain, between 0.01Hz and 10Hz).

The overall smartphone capabilities show reliable data for train-track interaction monitoring. Further information can be found in the relevant published article [6].

In conclusion, the research led by SNCF Réseau clearly shows that for the purpose of measuring acceleration inside a train carbody for track maintenance goals, a smartphone is comparable to the standard reference system.

5.4. RIDE COMFORT ANALYSIS WITH SMARTPHONES

The University of Zagreb has tested and compared data acquired by a Samsung Galaxy S8+, a Samsung Galaxy A52s and an acquisition system created at their university (cDAQ). The two smartphones were using a commercial application to save and then use the data, which was collected on a tram inside a dense city area. The three measurement features were mounted on a steel brick to have the same transfer function and eliminate the smartphone covers having an impact (see Figure 17).

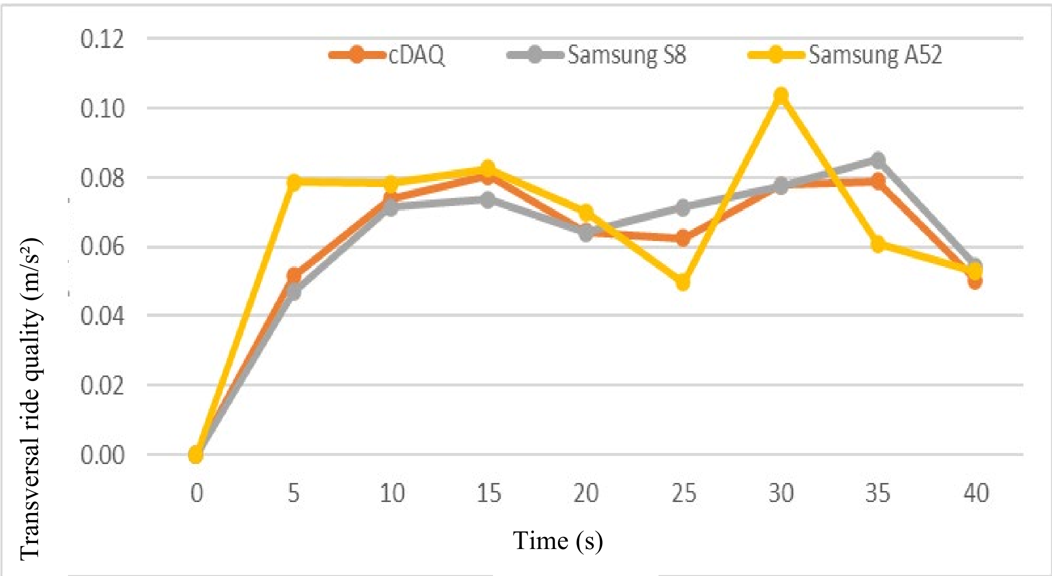


Figure 17: Comparisons of the ride quality - obtained by the smartphones and reference system

The standards from *EN 12299* on ride comfort quality were assessed with the three measurements systems and compared with each other.

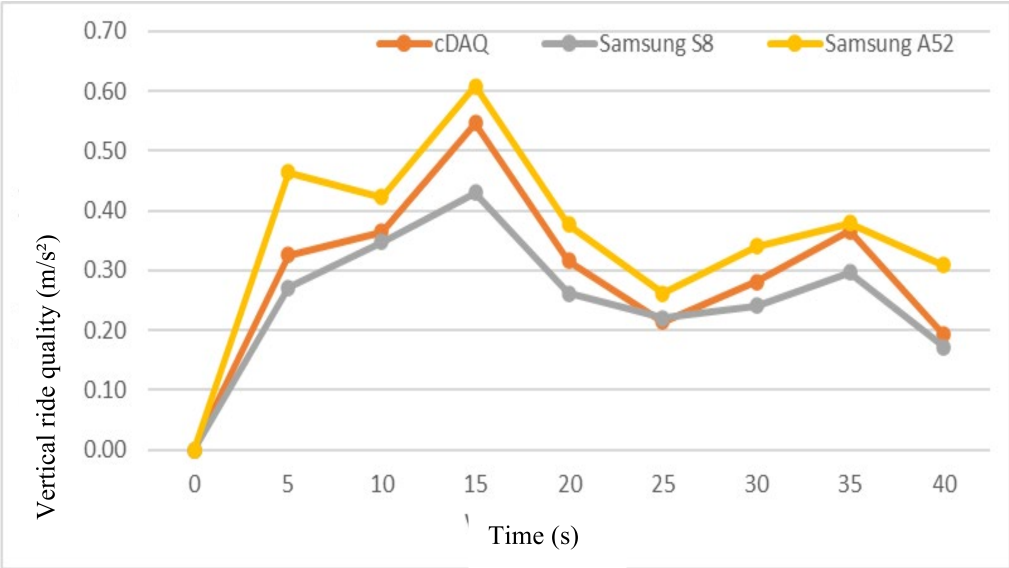


Figure 18: Comparisons of the ride quality - obtained by the smartphones and reference system

In Figure 18, the curves obtained show similar results between the different systems, with the Samsung A52s showing both higher values in general and more differences than between the Samsung S8 and the reference system. Yet, this observation shows that smartphones can be used for comfort assessment on train lines.

5.5. SUMMARY

The work conducted in this chapter showed how certain smartphones can be used for train acceleration measurements. SNCF Réseau and the University of Zagreb have explored two different approaches to assess the quality of different smartphone measurements. Smartphones measurement quality should be evaluated taking into account the specific use cases of each infrastructure manager, depending on train types, speeds and track quality.

6. IMPACT OF THE ACCELEROMETER POSITION

The aim of this chapter is to give precise guidelines on the use of accelerometers inside train carbodies by defining the location that should be used to observe the most relevant accelerations. Two different approaches can be used to this effect:

- Taking measurements at the most critical location
- Taking measurements at the most relevant location to assess track quality

These two positions are not always the same and it is up to the infrastructure manager to choose which one is relevant to them.

6.1. LONGITUDINAL POSITION ALONG A TRAIN

Different test campaigns were organised in 2021 and 2022 to measure acceleration inside IRIS 320. The same smartphone model (iPhone 8) was positioned at different locations inside the train in order to compare the measured accelerations.

The smartphone's positions to compare the longitudinal position are given in Figure 19. One limitation of this test was that the front (or head) and rear (also tail or back) positions were not located at the extreme end of the train, but in between two coaches at approximately 40m from the nose.

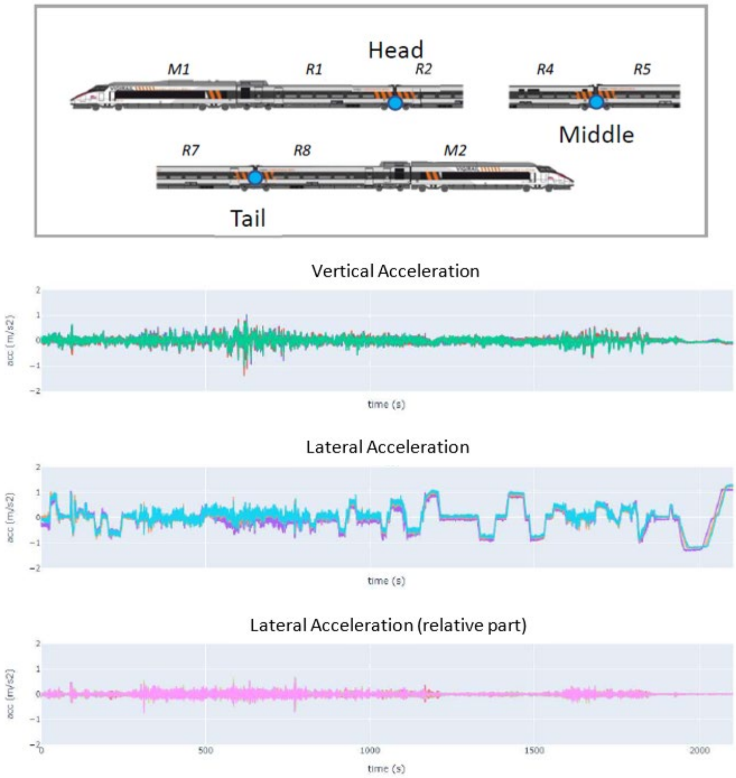


Figure 19: Graphic comparison of the front, middle and rear accelerometers

Measurement points 1: R1/R2, 2: R4/R5, 3: R7/R8

The comparison on the graph between the different measuring points shows very similar results for the vertical and lateral accelerations. Nevertheless, the differences are higher in the lateral direction, as shown in Table 2.

Table 2: C95 of the difference between measurement points

95% of the difference:	Vertical acceleration	Lateral acceleration	Lateral acceleration (high frequencies)
Front – middle	0.20	0.26	0.14
Front – rear	0.25	0.32	0.16
Middle - rear	0.20	0.15	0.11

In order to assess the impact of the longitudinal position in a coach, the maximum acceleration and the RMS values of the signals have been compared. The maximum acceleration was taken as 99.85% of the distribution of the dataset. The results are illustrated in Figure 20.

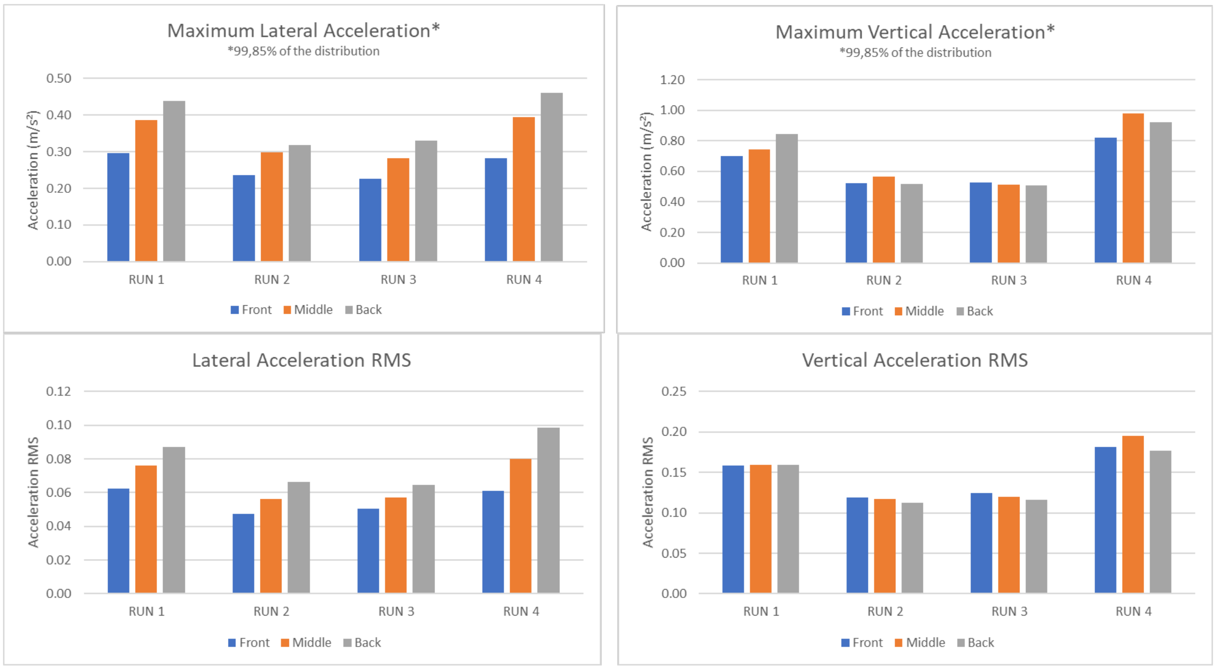


Figure 20: Comparison of the maximum values of the lateral and vertical accelerations

There is an increase in the maximum observed values in the lateral direction towards the rear of the train with a +35 to +50% difference between the front and rear. The RMS value also increases by +30 to +60% when comparing front and rear values.

The effect of the longitudinal position is less clear on the vertical acceleration. The values observed show no clear trend and the position has a low impact.

6.2. LONGITUDINAL POSITION ALONG A COACH

Another test conducted by the SNCF Réseau team was to measure accelerations in three different longitudinal positions along a coach (see Figure 21).

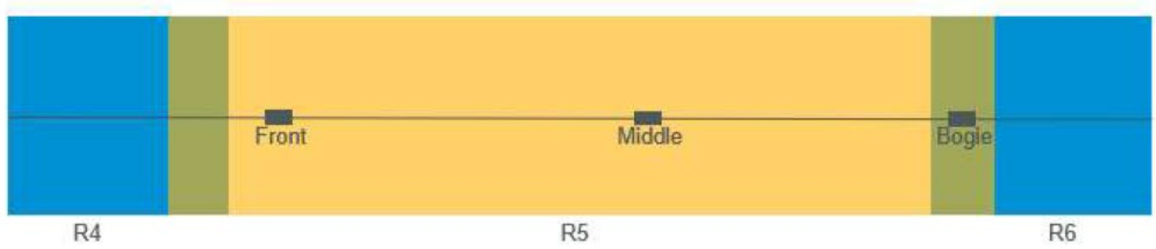


Figure 21: Positions of a smartphone aboard coach 5

The smartphone named “Front” was located towards the front of the coach but not on top of the bogie. The smartphone named “Middle” was located in the middle of the coach. And the smartphone named “Bogie” was located at the back of the same coach, on top of the bogie.

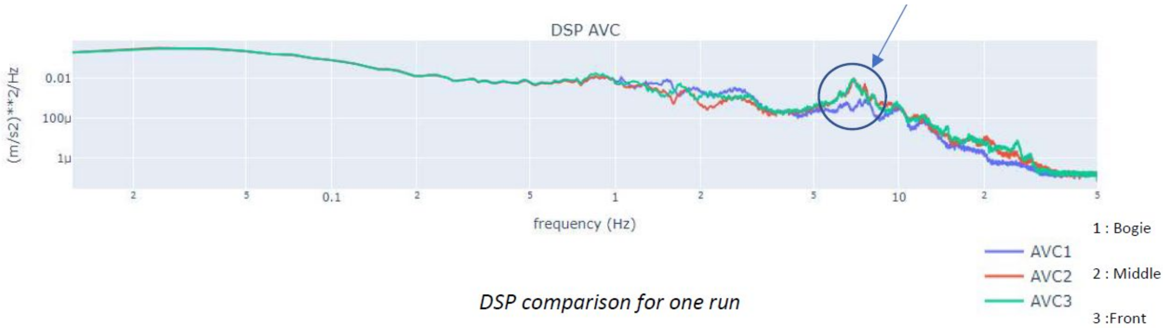


Figure 22: PSD comparison of the vertical acceleration – varying longitudinal positions

Figure 22 give a comparison of the PSD for the signals, which shows a peak of approximately 7Hz in the vertical acceleration, but only on the middle and front accelerometers. A hypothesis is that this peak corresponds to the bending frequency of the coach, but this is not observed on the bogie as this latest is near a node of the coach bending mode.

The front and middle measurements gave very similar results, which was expected. Higher lateral accelerations were observed on top of the bogie (+15%), whereas higher vertical accelerations were observed in the middle of the carbody (+9%). This is illustrated in Figure 23, where the maximum acceleration and RMS values at each position inside the coach are compared.

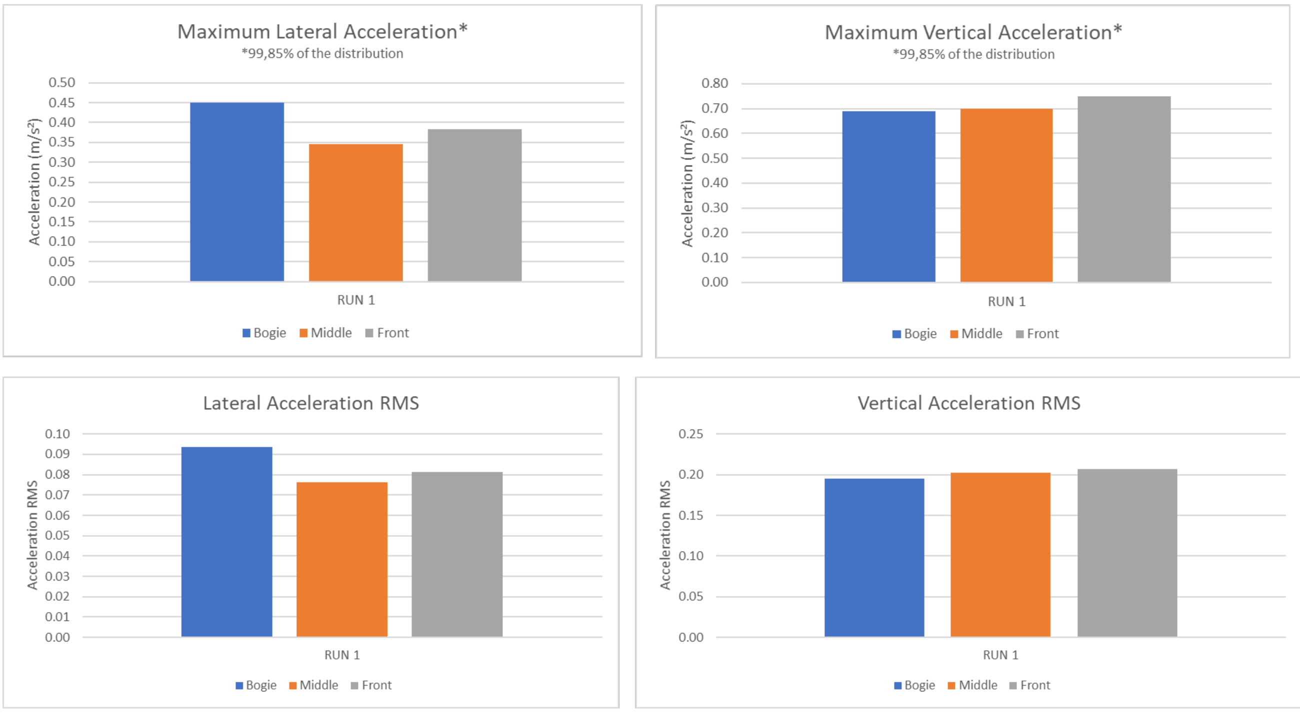


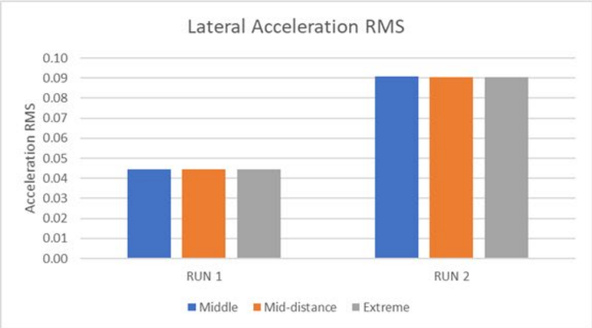
Figure 23: Comparison of the maximum values of the lateral and vertical accelerations

6.3. LATERAL POSITION

The next comparison took place in the same coach with a spacing of 400 millimetres in the lateral direction between each smartphone. One was placed at the centre of the coach, and the two others to the left (in the direction of travel), as shown in Figure 24.

Figure 24: Smartphone setup for comparing the lateral position

The same method as given in 6.2 was used to compare the results. The maximum acceleration and RMS values for the different lateral positions were close to identical as seen in Figure 25. Thus, the lateral position of the smartphone is not considered notable. However, the smartphone should be kept within a reasonable distance from the middle axis of the train to avoid a bias in the detection of defects from both rails. This distance can be set to 400mm.



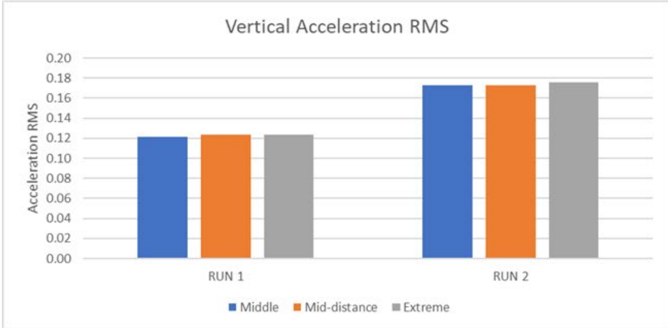


Figure 25: Comparison of the RMS values of the lateral and vertical accelerations

6.4. SUMMARY

The aim of this chapter was to provide guidelines for choosing the measurement position inside rolling stock. The most critical position to evaluate the lateral acceleration is towards the rear of the train, and to assess track quality, it is recommended that the accelerometers be placed above the bogie and along the longitudinal axis of the train.

CONCLUSION

This report has shown the types of low-cost devices that can be used to monitor tracks with onboard accelerometers. Different setups can be used, from sensor and development board combinations to smartphones. The methods for collecting data have been described and the precision of different smartphones evaluated, with similar results being obtained between the reference systems and smartphones. This, therefore, proves their usefulness for future deployment in the track maintenance domain.

Different methods were explored for pinpointing the location of the measured data, which can be applied to smartphones as well as conventional accelerometer setups onboard trains, even though smartphones rely heavily on GPS locations. Finally, different studies were conducted to evaluate the impact on measurements of the device's position onboard a train. While, by definition, portable devices can be placed anywhere, observations from the tests showed that the measuring system should be placed at the rear of the train, on top of a bogie, and if possible, centred in the lateral direction, as this would allow for the highest level of peak detection, as well as indirect track defect observation through the carbody's movements.

As there is an increasing interest from universities and infrastructure managers in having large databases with measurements coming in on a daily or even hourly basis using low-cost and connected devices, this SWG has shown that smartphones and low-cost devices can answer this need by constituting cheap pieces of measuring equipment that can be easily installed in a commercial train.

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APPENDIX 1: RUMO LOGISTICA'S DEVICE

The initial measurements that Rumo logistica took and the results obtained are presented below as well as a short overview of the setup.

Figure 26: General overview of the Micro controller, SD storage, and GPS

The initial measurements were carried out along 120 kilometres of the railway track that connects the city of Curitiba in the state of Paraná to the city of Morretes on the coast. Figure 27 shows the route of the section in yellow. Sensors were placed on the floor of a locomotive to carry out the measurements.

Figure 27: First section measured – Rumo Logistica

Figure 28 shows the data received from the acceleration sensor along the X, Y and Z axes. This data was measured with georeferencing and was then correlated with the data from the measurement car, to define the limit values of surface deviations, misalignments, and other operational risks.

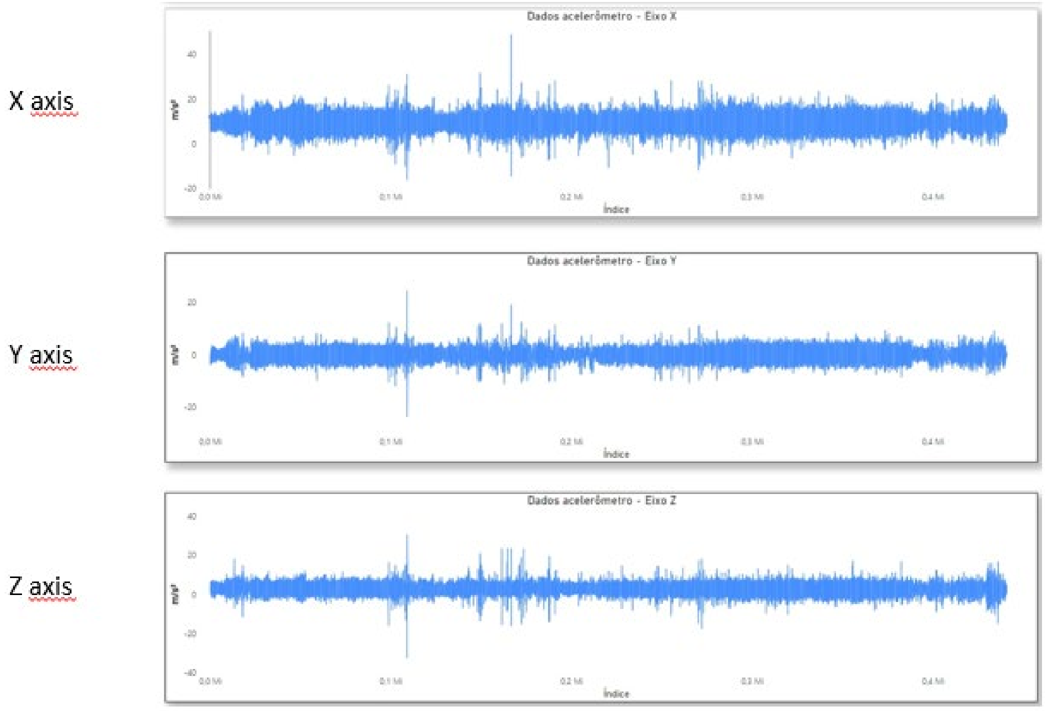
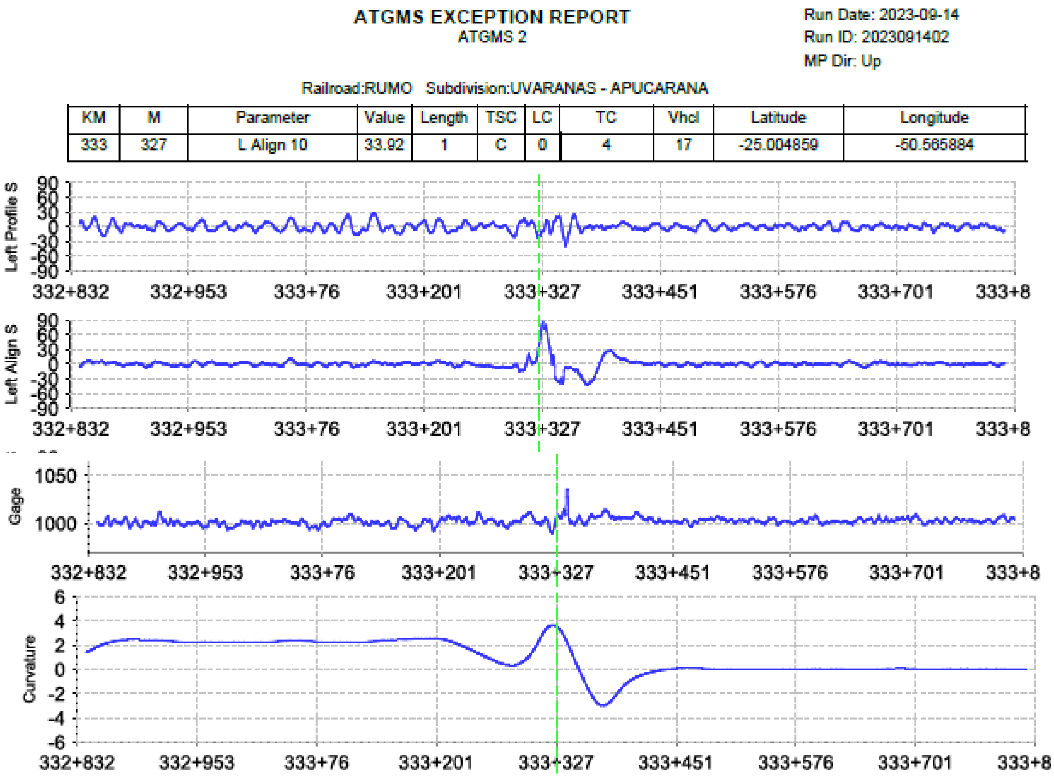


Figure 28: Acceleration data – Rumo Logistica

Figure 29, shows the data from the measurement car, which was used to create the correlation between the accelerations detected by the sensors and the deviations found on the track.


ATGMS EXCEPTION REPORT
ATGMS 2

Run Date: 2023-09-14
Run ID: 2023091402
MP Dir: Up

Railroad:RUMO Subdivision:UVARANAS - APUCARANA

KM	M	Parameter	Value	Length	TSC	LC	TC	Vhcl	Latitude	Longitude
333	327	L Align 10	33.92	1	C	0	4	17	-25.004859	-50.565884

Figure 29 : Data from a measurement car – Rumo Logistica

APPENDIX 2: DATA REFERENCES

A.2.1. IRIS 320

The first reference system used at SNCF Réseau was the IRIS320 diagnostic train, currently in use on the network for track geometry quality diagnostics, as well as to measure the dynamic interaction between trains and infrastructure. It circulates on high-speed lines operating at a speed of 300km/h.

Figure 30 - Iris320 Diagnostic Vehicle

Train setup

Vertical and lateral in bogie (AVB-ATB) and in carbody (AVC-ATC) acceleration signals are used.

The bogie accelerometer is located near the centre of the bogie, without maintaining the double arrangement on the right and left as is usually the case for axle box accelerometers.

On the other hand, the accelerometer in the carbody is located between the car bodies, at floor level and aligned with the accelerometer on the bogie.

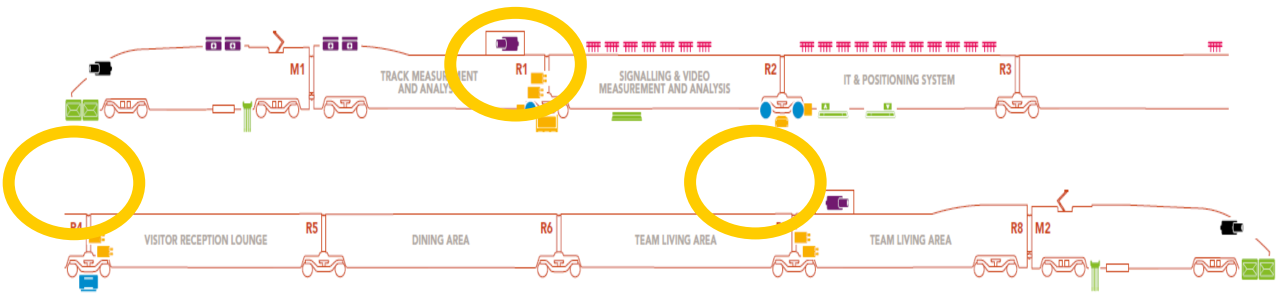


Figure 31: Diagram of the Iris 320 accelerometer setup

A.2.2 PORTABLE ACCELEROMETER SYSTEM (ACCELEROMETRIC SUITCASE)

The portable accelerometric systems are composed of 3 main parts:

- Accelerometer
- GPS sensor
- Acquisition system (PC, live processing box, battery)

Figure 32 and Figure 33 provide an overview of the system, which is mainly used on conventional lines although it can also be used on high-speed lines.

Figure 32 : Accelerometric and GPS sensors

Figure 33 : Acquisition system

A.2.3. PIRI REIS

The Piri Reis diagnostic train, currently in use on the Turkish network for track geometry quality diagnostics, as well as to measure the dynamic interaction between trains and infrastructure, is another reference system.

Figure 34: PiriReis diagnostic vehicle

Train setup

The Piri Reis diagnostic train is equipped with single-axis accelerometers in the axle boxes, bogies and carbodies, and arranged in the vertical and transversal direction.

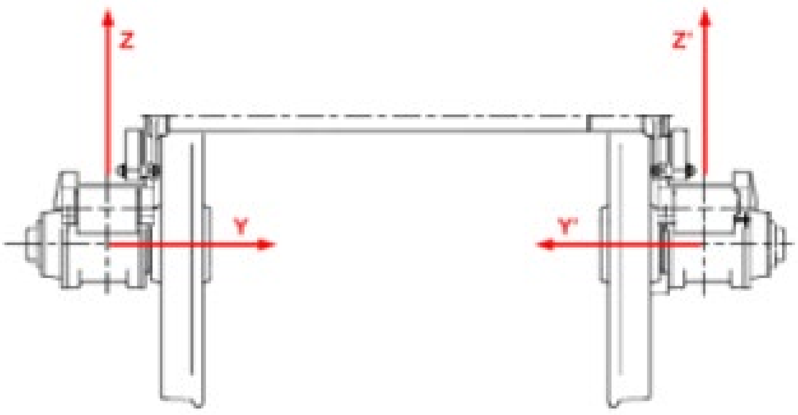


Figure 35: Accelerometer displacement

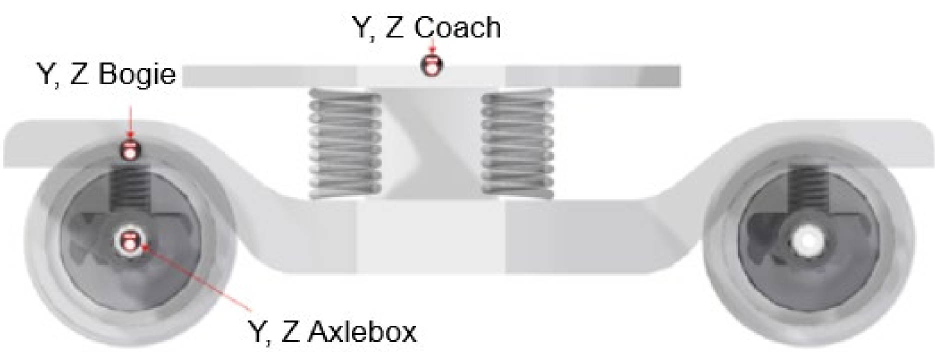


Figure 36: Accelerometer positions

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